Scalability & Future Plan

Date	15 March 2026
Team ID	SB25-PNA-1042
Project Name	Personalized Networking Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current Personalized Networking Assistant solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority
1	Local JSON files (history.json, feedback.json) used for storage	Limits concurrent access and does not scale beyond a single-user local deployment.	High
2	Single-instance GPT-2 Small model running on CPU	Response latency increases under multiple simultaneous generation requests.	Medium
3	No user authentication or multi-user session management	Prevents the assistant from supporting multiple users with separate histories.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local user via Streamlit session	Migrate to a multi-user web deployment with session-based authentication and load balancing.
2	Data Storage	Local JSON files for history and feedback	Move to a managed database (PostgreSQL) for structured, concurrent, and queryable storage.
3	Performance	Synchronous GPT-2 inference on CPU	Deploy quantized/optimized models on GPU-backed inference endpoints with response caching.
4	Security	No authentication; local-only access	Introduce OAuth-based login, API key management, and encrypted storage of user data.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Migrate storage from JSON files to PostgreSQL	1-2 months	Reliable, concurrent, multi-user data persistence.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add user authentication and personalized profiles	2-3 months	Enables individualized conversation history and recommendations.
Phase 3	Upgrade to larger fine-tuned LLM with GPU inference	3-4 months	Higher-quality, faster conversation starter generation.
Phase 4	Deploy as a cloud-hosted, mobile-responsive application	5-6 months	Broader accessibility and support for larger networking events.